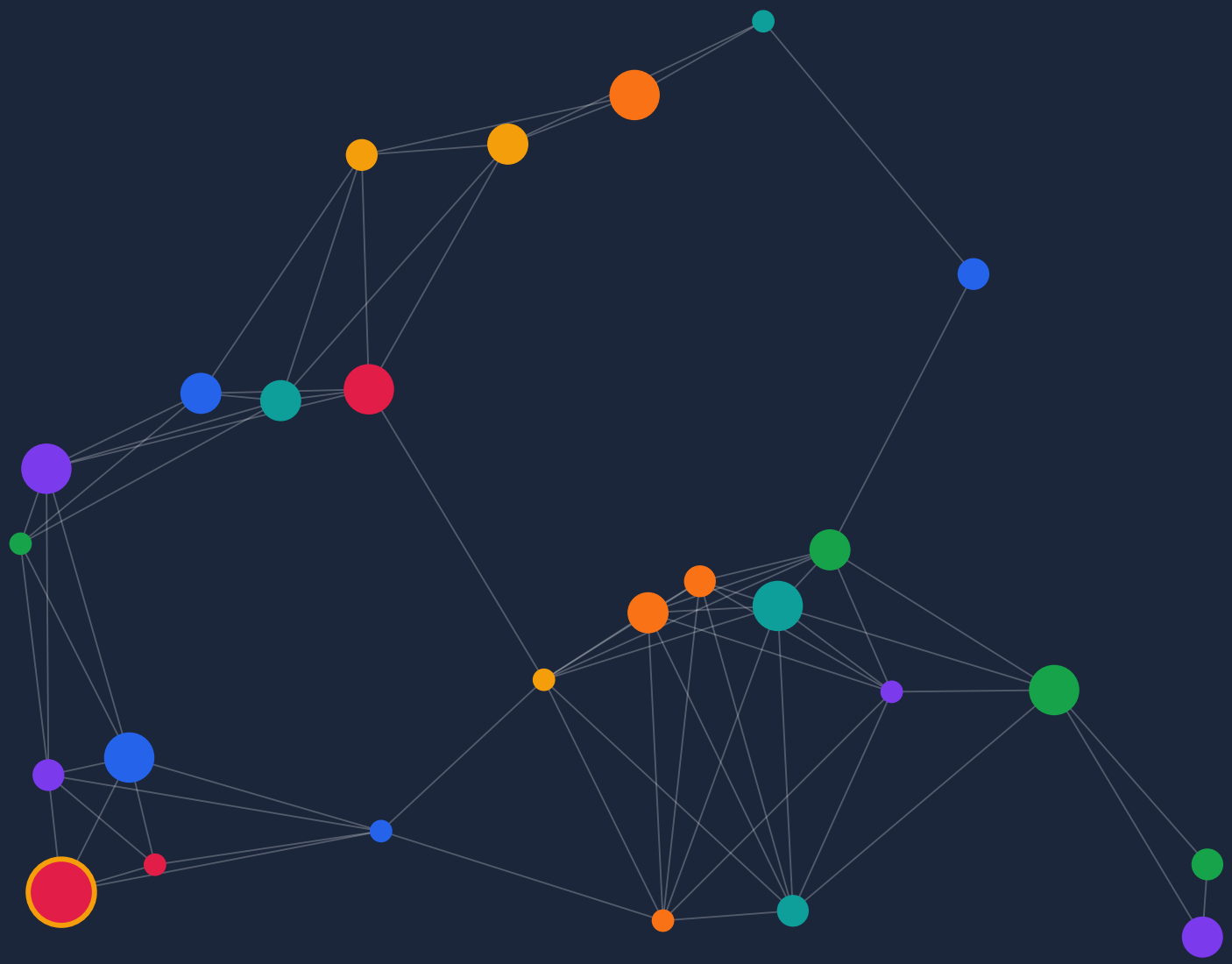


SMART INDIA HACKATHON 2026 | PS SIH26189 | TEAM UNITx

Case Connect AI

AI-Powered Criminal Network Analysis System

The complete master guide: problem, solution, every feature,
how it is built, why each choice was made, and how it is different.

Written in plain English, so anyone can understand it,
even with zero technical or police background.

FIR | Call records | Bank data | Graph | AI | Human decides

All people, accounts and numbers in this document are fictional examples created for explanation.
Draft prepared October 2026.

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How to read this guide

If you are...	Read these parts	Why
A complete beginner	Quick Summary, Part 1, 2 and 3	Understand the problem and the idea through a simple story.
A developer on the team	Parts 4, 5, 7 and 8	Features, tools, build order and what to learn.
The presenter / pitch lead	Parts 3, 6, 9 and 10	Story, uniqueness, demo script, judge questions and honest limits.
Anyone unsure about a word	Part 12 (Glossary)	Every technical word explained in one line.

Colour tags used in this guide

HERO FEATURE | FOUNDATION | SUPPORT | EXTRA | ROADMAP

- **Hero feature:** the 3 features that make the project stand out. Build these best.
- **Foundation:** must exist for everything else to work (login, cases, upload, graph).
- **Support:** makes the product complete and trustworthy (timeline, report, audit log).
- **Extra:** small additions we suggest that cost little but add value.
- **Roadmap:** good ideas for later; do not build them first.

> **FICTIONAL EXAMPLES**
> Every name, number and account in this document is **fictional**. The example case (Case #101) is used again and again, so you can follow one story from the start to the end.

After reading this guide you will be able to...

- explain the **problem** and the **idea** to anyone in two minutes;
- name every **feature**, say **why** it exists and **how** it works;
- say which **tool** is used for what, and why it beat the alternatives;
- explain **how we are different** from existing products, honestly;
- follow a **step-by-step build plan** and know what to learn first;
- run the **demo** and answer the **judges' questions** with confidence.



The whole system in seven steps (explained in Part 2).

The Project in 60 Seconds

If you read only one page, read this one

Case Connect AI is a website for police investigators. An officer uploads the evidence of a case: FIR reports, phone call records and bank statements. The system reads the files by itself, joins everything into **one connected map** (called a *graph*) and shows **who is connected to whom**, **where the money went**, and **which people deserve a closer look and why**. Every fact links back to its source file. **The officer, not the AI, makes the final decision.**

THE PROBLEM	THE SOLUTION	THE RESULT
Evidence is scattered across files and police stations. Criminals hide behind layers of middlemen. Officers join the dots by hand in Excel. That is slow, and it gets harder with every extra step.	AI reads documents and files. Duplicate identities are matched (officer approves). Everything goes into a graph database. The system analyses money and calls and explains every flag with evidence.	One connected picture per case and across cases. Hidden people and money paths become visible. Every claim is traceable to its source. A ready-made PDF report for seniors and courts.

Our 3 hero features

1 Master Profile	2 Money Trail Explorer	3 Lead Advisor
Same person in many FIRs (even different police stations)? The system suggests the link; the officer approves; one profile shows everything together.	Pick an account, set an amount filter (for example ₹5 lakh and above) and a hop limit. See where the money came from and where it moved.	Click a person and see *why* the system flagged them: the connections, transactions, evidence, what data is still missing and what to do next.

Our 4 golden rules

1	**AI suggests, the human decides.** Merges, flags and leads are always shown as *suggestions* with a confidence level.
2	**Every fact has a source.** Click any link and see the file, page or row it came from.
3	**Every flag is explained.** No black box: scores come from simple, visible calculations.
4	**Start small and real.** A smooth demo on synthetic data beats 20 half-working features.

The example case used throughout this guide (Case #101, fictional)

4	₹12 lakh	3	6	1
victims	lost in total	FIRs, 3 stations	people in the network	hidden organiser found

1 The Problem

Why investigators struggle today

A simple way to picture it

Imagine a jigsaw puzzle whose pieces are kept in **five different rooms**, in five different formats. Some pieces belong to the same picture but have **different names** written on them. Nobody can see the whole picture. That is how evidence in organised-crime cases looks today.

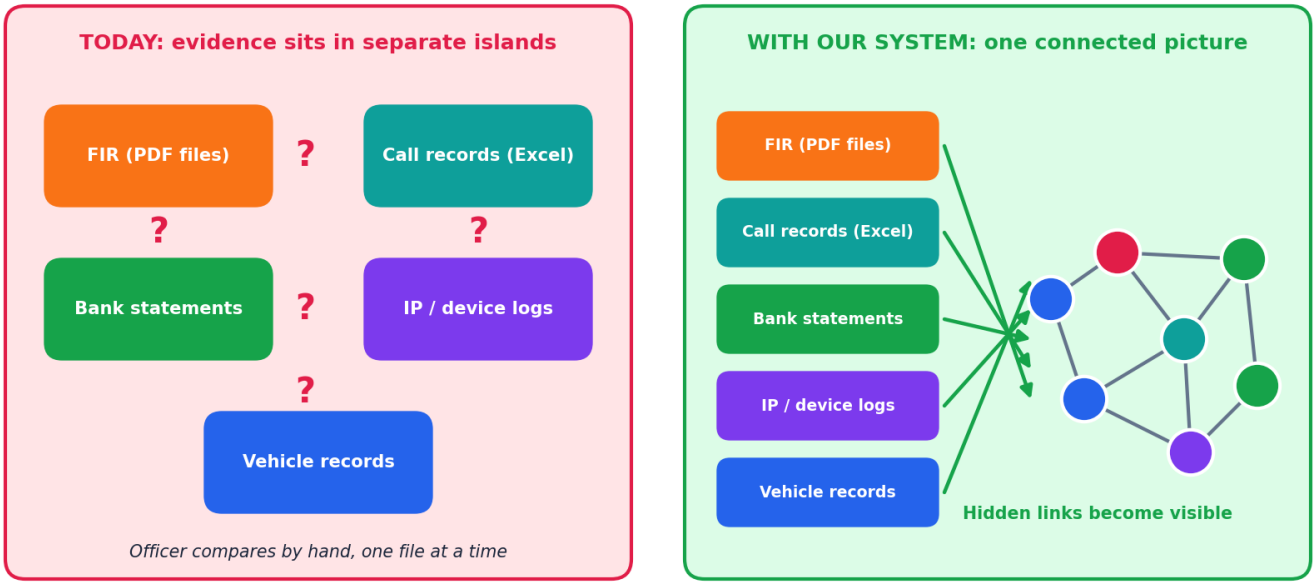


Figure 1: Today the officer joins separate files by hand. With the system, all evidence feeds one connected picture.

What kinds of crime are we talking about?

- **Cyber fraud** and online investment scams, where money is split across many bank accounts.
- **Money laundering** and hawala-style networks, where cash moves through middlemen and shell companies.
- **Trafficking and drug networks**, where members talk through burner phones and keep their distance from the leader.

The 3 root causes

1 Scattered evidence	2 Hidden middlemen	3 Same person, many names
FIRs sit in PDFs at police stations. Call records (CDRs) come as Excel or CSV files from telecom companies. Bank statements arrive in different formats. There is no common format, so nothing connects automatically.	The real organiser stays several steps away from the crime, working through proxy accounts and shell companies. Each step is called a **hop**. By hand, tracking stops being practical after about 2 to 3 hops.	"Rahul Sharma", "R. Sharma" and "Rahul K. Sharma" may be one person. In a large city, the same person can appear in FIRs at many police stations, and each file holds only a small piece.

What this costs

Officers clean call records and bank statements in Excel by hand. This can take days or even weeks for one gang (this is an estimate from the problem statement analysis, not a measured number). Meanwhile, criminals move the money out and change phones. **Speed matters: the longer it takes, the less money can be frozen.**

Who is affected?

Who	Their pain today
Investigating officers (IOs)	Hours of manual cleaning and cross-checking before real investigation even starts.
Cyber crime cells and state police intelligence wings	Many cases, many phones and accounts, little time; hard to see links between cases.
Central agencies (CBI, ED, NIA)	Need links across states and across agencies; evidence arrives in different formats.
Prosecutors and courts	Need evidence whose origin can be proven and shown step by step.
Victims	Wait longer; stolen money has often moved on before action is taken.

WHERE WE FIT (CCTNS)

India already has **CCTNS** (Crime and Criminal Tracking Network and Systems), a national network that connects police stations for FIR and crime-record management. The Ministry of Home Affairs told Parliament that all **17,762 police stations** were using it as of 1 June 2026. Case Connect AI does **not** replace CCTNS. It is an **analysis layer**: it reads call and bank evidence, connects it with FIR data and shows hidden links. Real integration with official systems would need formal approval and authorised access.

2 Our Solution

One connected picture, built with AI, controlled by humans

In one sentence: **we turn scattered evidence into one connected map, and we explain what stands out on it.** The pipeline below is what happens every time an officer uploads files.



Figure 2: The 7-step pipeline. Two points need a human: reviewing matches (step 4) and the final decision.

Our 5 promises

1. **See everything in one place**
   Upload FIRs, call records and bank statements into one case workspace.
2. **Find hidden links**
   A graph shows connections 1 to 4 hops away, and links the same person across different FIRs.
3. **Follow the money**
   An interactive money trail with amount and hop filters shows where money came from and moved.
4. **Explain every flag**
   "Why is this person important?" lists the connections, transactions and evidence behind a flag.
5. **Keep humans in charge and evidence safe**
   Officer approvals, SHA-256 file fingerprints, roles and an audit log protect the process.

Today versus with Case Connect AI

Task	Today (manual)	With our system
Read FIRs	Officer reads and types names and numbers into Excel.	AI extracts names, phones, accounts, amounts. Officer only checks.
Combine FIR + call + bank data	Copy-paste and lookups across many sheets.	Automatic: everything becomes one graph.
Find the hidden middleman	Follow a few links by eye; gets hard after 2 to 3 hops.	Expand 1 to 4 hops in one click; path highlighted.
Spot the same person in other FIRs	Depends on memory or phone calls between police stations.	System suggests the match with confidence; officer approves.
Trace money	Scroll through long bank statements.	Money Trail Explorer with amount and hop filters.
Explain to a senior or court	Write notes by hand.	One-click PDF report with evidence references and file fingerprints.

WHAT THE SYSTEM WILL NEVER DO

- Declare anyone a criminal. It only shows *leads for review*.
- Merge two identities without an officer's approval.
- Hide where a fact came from.
- Take any legal or enforcement action.

3 A Day with Inspector Meera

One fictional case, told step by step

Case #101 (fictional): an online investment fraud. Four victims were tricked into paying ₹3 lakh each, ₹12 lakh in total. Three police stations received three separate FIRs. Inspector Meera Rao has the FIR PDFs, a call-records file and a bank-statement file. Here is how she works with Case Connect AI.

#	What Meera does	What the website does
1	Logs in as an **Investigator**.	Checks her role and records the login in the audit log.
2	Creates "Case #101 - Online Investment Fraud".	Makes a private workspace. Only assigned officers can open it.
3	Uploads 3 FIR PDFs, a call-records CSV and a bank CSV.	Checks file type and columns. Creates a SHA-256 fingerprint for every file. Keeps originals untouched.
4	Waits a few seconds.	AI reads the files and extracts names, phones, accounts, amounts, vehicles and dates, each with a confidence level and source page.
5	Sees a suggestion: "Rahul Sharma" and "R. Sharma" may be the same person (HIGH: same phone). She clicks **Approve**.	Merges them into one Master Profile. Logs who approved and when.
6	Opens the Investigation page.	Draws the graph: people, phones, accounts, vehicles and the calls and transfers between them.
7	Searches "Rahul Sharma" and chooses **3 hops**.	Shows the chain Rahul to Amit to Vikas to Shree Trade Links. A new name appears: **Vikas Yadav**.
8	Opens the Money Trail: Account A, filter ₹3 lakh and above, 3 hops.	Shows ₹12 lakh entering Account A, splitting into B, C and D, joining again in Account E (Vikas), and ₹10 lakh leaving to a shell company.
9	Clicks **Vikas Yadav**.	Lead Advisor panel: priority 83/100, the reasons, evidence links, missing data and suggested next steps.
10	Clicks **Generate report**.	Creates a PDF with entities, graph picture, money trail, timeline, evidence references and file fingerprints.

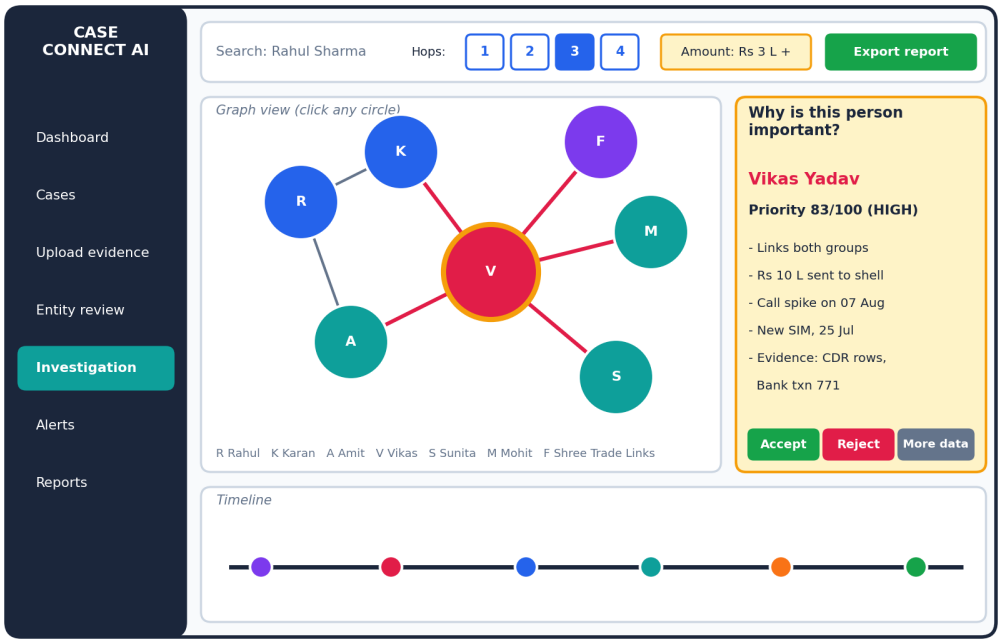


Figure 3: A simple sketch of the Investigation screen: search and filters on top, graph in the middle, the 'Why is this person important?' panel on the right, timeline at the bottom.



Figure 4: What really happened in Case #101, in time order. The system sees the pattern before any FIR is filed.

Fictional example: who called whom (line thickness = number of calls)

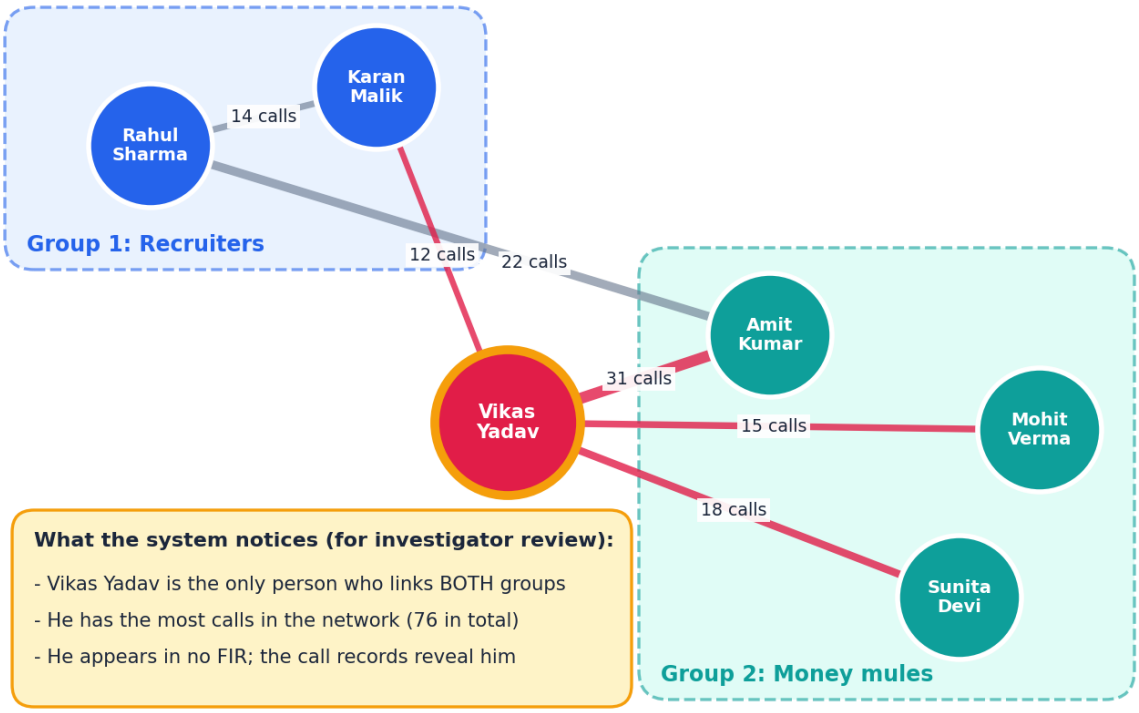


Figure 5: Who called whom. Vikas Yadav is the only person who links both groups.

THE KEY INSIGHT

Vikas Yadav is not named in any of the three FIRs. He is found only because the system connects the call records and the bank data. This is the 'aha moment' of the whole project, and the moment to show first in a demo.

4 Features in Detail

What each feature does, why it exists, how it works and why we chose it

There are **14 core features** plus **5 small extras**. Each feature has a card with four rows: **What it does** (in plain words), **Why it exists**, **How it works** and **Why this choice** (including the alternatives we rejected).

#	Feature	Type	One-line summary
F1	Secure login and roles	**FOUNDATION**	Right people see the right things
F2	Case workspace	**FOUNDATION**	Each investigation in its own protected space
F3	Evidence upload and fingerprint	**FOUNDATION**	Upload files; prove they were not changed
F4	AI evidence extraction	**FOUNDATION**	Reads files and finds people, phones, accounts, amounts
F5	Entity matching and review	**FOUNDATION**	Finds duplicate identities; officer approves
F6	Cross-FIR Master Profile	**HERO**	One profile from many FIRs and stations
F7	Knowledge graph and multi-hop search	**FOUNDATION**	See who is connected, 1 to 4 hops away
F8	Money Trail Explorer	**HERO**	Follow money with amount and hop filters
F9	Communication pattern detection	**SUPPORT**	Spikes, bridges, sudden new links
F10	Lead Advisor: 'Why is this person important?'	**HERO**	Explained leads, missing data, next steps
F11	Timeline	**SUPPORT**	Everything in time order
F12	'Why this link?' evidence button	**SUPPORT**	Every link shows its source
F13	Audit log with tamper check	**SUPPORT**	Who did what, protected from edits
F14	Report generator	**SUPPORT**	One-click PDF with evidence references
F15	Five small extras	**EXTRA**	Cross-case alert, data check, decision log, saved views, summaries

F1 Secure login and roles — FOUNDATION

What it does	Officers sign in with a user name and password. Each role sees and does only what it should: **Admin**, **Investigator** or **Viewer**.
Why it exists	Case data is highly sensitive. Roles stop the wrong person from seeing or changing evidence.
How it works	Passwords are stored as **hashes** (bcrypt), never as plain text. After login the server gives a signed token (JWT). Every request is checked against the user's role.
Why this choice	Standard, well tested and quick to build. Alternatives: single sign-on with government identity systems (roadmap), or writing your own security code (never do that).

Action	Admin	Investigator	Viewer
Upload evidence	**Yes**	**Yes**	**No**
Approve or reject entity matches	**Yes**	**Yes**	**No**
View graph, money trail, timeline	**Yes**	**Yes**	**Yes**
Generate report	**Yes**	**Yes**	**No**
Manage users and view the audit log	**Yes**	**No**	**No**

F2 Case workspace — FOUNDATION

What it does	Every investigation lives inside a **Case**: title, description, priority, status, assigned officers, notes and the list of evidence.
Why it exists	Keeps investigations separate and gives each officer access only to the cases assigned to them.
How it works	Cases are stored in PostgreSQL. Every graph item in Neo4j carries its case number. Links between cases are shown only to officers who can open both cases.
Why this choice	One big shared pool would be simpler, but it risks leaking data between cases and officers.

F3 Evidence upload and fingerprint — FOUNDATION

What it does	Upload FIRs (PDF or text), call records (CSV or Excel) and bank statements (CSV or Excel). The system checks, fingerprints and stores each file.
Why it exists	In court you must show that evidence was not changed. A **SHA-256 fingerprint** changes completely if even one character in the file changes.
How it works	Check type, size and columns, then compute SHA-256 (Python *hashlib*), save the original file read-only, and store the hash, uploader and time. Anyone can re-check the fingerprint later.
Why this choice	SHA-256 is standard, free and fast. Storing every file on a blockchain would be slow and unnecessary.

LEGAL NOTE

India's rules for electronic evidence moved from Section 65B of the Indian Evidence Act to **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023**, which asks for a certificate for electronic records. Our fingerprints and audit log help prepare that paperwork. Whether evidence is admissible is decided by the court and lawyers, not by software.

F4 AI evidence extraction — FOUNDATION

What it does	Reads text and finds **people, phone numbers, bank accounts, UPI IDs, vehicles, IP addresses, amounts, dates, places and organisations**.
Why it exists	Typing these by hand is slow and error-prone, and it is the first thing that eats an officer's time.
How it works	1. **PDF to text** with PyMuPDF or pdfplumber. 2. **Rules (regex)** for things that always look the same: phone, account, amount, date, vehicle number, IP. 3. **AI (spaCy NER)** for names, places and organisations. 4. **Clean-up:** +91-98765 43210 and 9876543210 become one standard phone; "Rs 5 lakh", "₹5,00,000" and "500000 INR" all become 500000. 5. Every entity gets a **confidence level** and its **source** (file and page). Call-record and bank CSV files need no AI: they are read directly with pandas.
Why this choice	Rules plus NER are fast, free, run offline and are easy to explain. **Alternatives:** an LLM can read anything, but it may invent facts and may need a paid service. Training your own model needs thousands of labelled examples. A later option: use an LLM only as a helper, with every output checked against the source text. **Scanned PDFs** need OCR (Tesseract); this is phase 2.

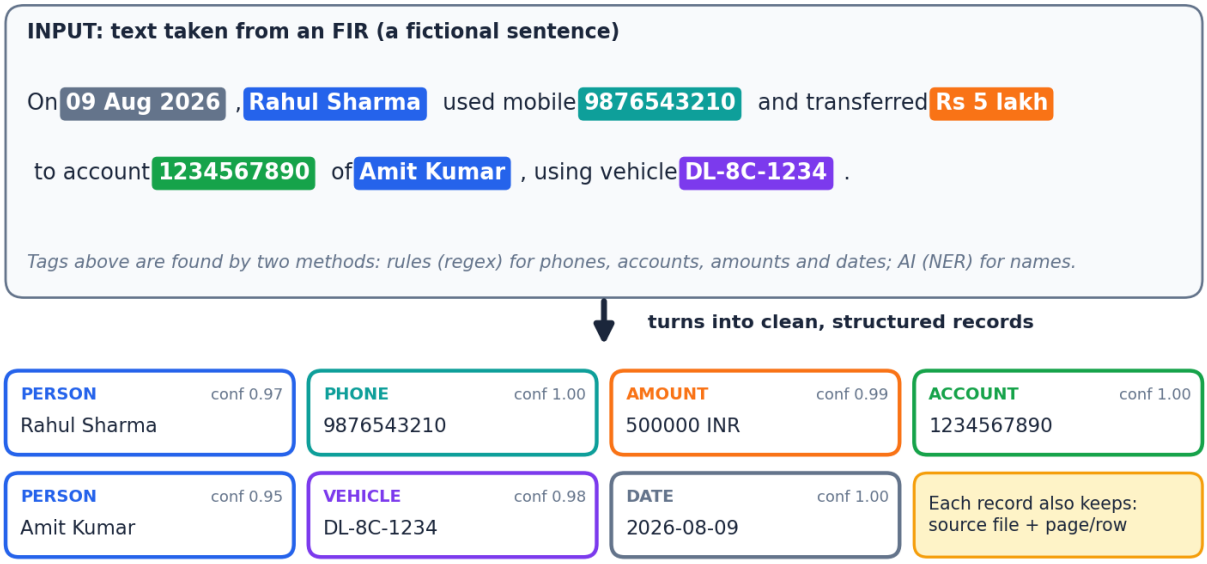
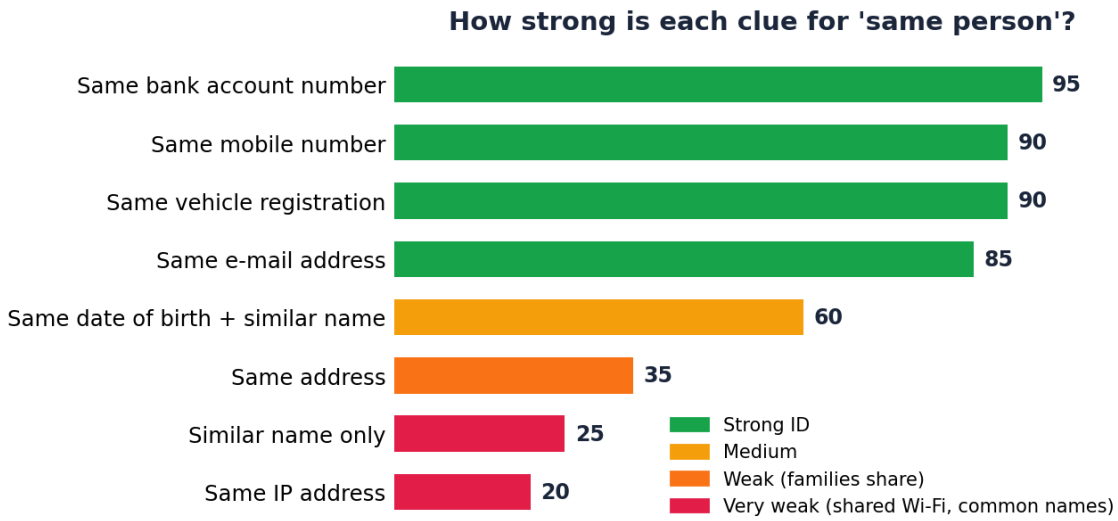


Figure 6: From a sentence in an FIR to clean, structured records (confidence values are illustrative).

F5 Entity matching and review — FOUNDATION

What it does	Finds records that may be the same person, account or phone and **suggests** merging them. The officer approves or rejects. Any merge can be undone.
Why it exists	One person appears under many spellings. A wrong merge sends an investigation in the wrong direction, so the system must be careful.
How it works	1. Exact match on **strong IDs**: bank account, phone, vehicle number, e-mail. 2. **Name similarity** with the Jaro-Winkler method (Python library rapidfuzz). 3. Shared details such as address and date of birth. 4. Combine into a confidence level: **HIGH** = a strong ID plus a similar name; **MEDIUM** = two medium clues; **LOW** = name only (shown as 'possible', never as a merge suggestion). 5. **Families and offices:** if one phone or address belongs to several different names, flag it as 'shared device/address' instead of merging.
Why this choice	Clear rules can be explained in one sentence to a judge or a court. A machine-learning matcher needs labelled data we do not have. We never auto-merge: this matches our promise of human control.



Illustrative strengths (0-100) that the team can tune. Merge always needs officer approval.

Figure 7: Some clues are strong (an account number) and some are weak (a common name or a shared Wi-Fi). Strengths are illustrative and can be tuned.

F6 Cross-FIR Master Profile

HERO FEATURE

What it does	Builds **one page per real-world person** with everything known from all FIRs and files: other names, phones, accounts, vehicles, cases, connections and timeline.
Why it exists	In a big city like Delhi the same person can appear in many FIRs at different police stations. Each FIR holds a small piece: a phone here, an account there. Joining the pieces reveals the real pattern.
How it works	When a new FIR is uploaded, the system looks for matching strong IDs in existing records, suggests links, and asks the officer. After approval the pieces join one Master Profile. Every fact keeps its FIR number and page.
Why this choice	Today officers search other stations manually or by phone. Safety rules: suggest links only for cases the officer may open, and always offer 'Undo merge'.

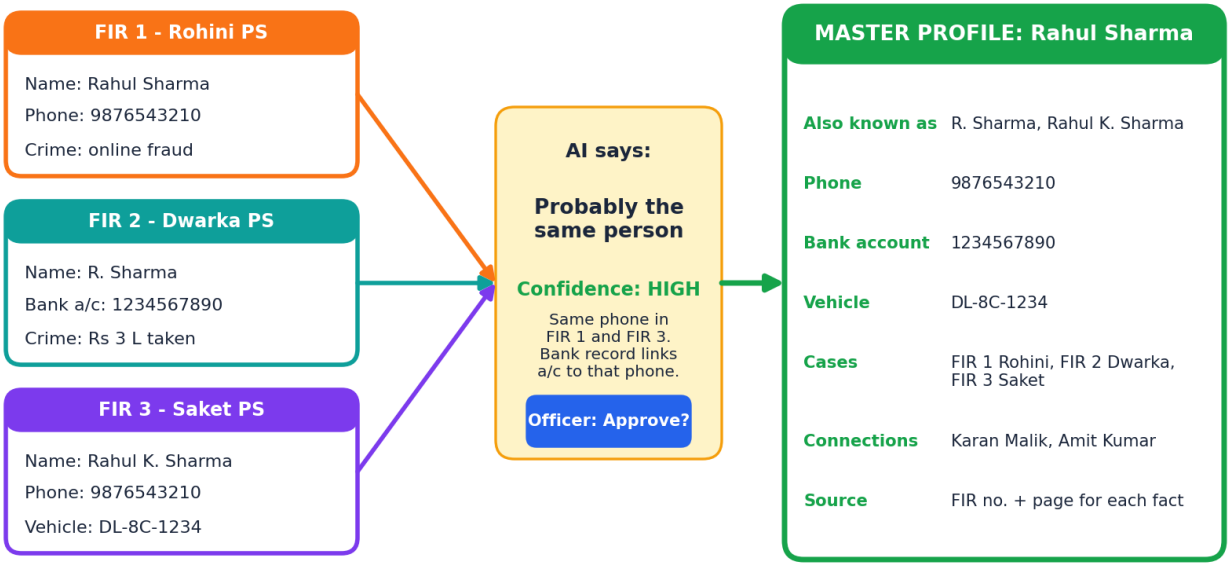


Figure 8: Three FIRs from three police stations become one Master Profile after the officer approves. The bank record, which lists 9876543210 as the registered mobile of the account, connects FIR 2 to the others.

F7 Knowledge graph and multi-hop search

FOUNDATION

What it does	Stores all entities and links as a **graph**. Search by name, phone, account, vehicle, IP or FIR. Expand **1 to 4 hops**. Filter by type, date and amount.
Why it exists	The question an investigator asks is: *who is connected to whom, and through whom?* A graph database answers this directly.
How it works	**Neo4j** stores nodes (Person, Phone, BankAccount, Vehicle, IP, FIR) and relationships (USES, OWNS, CALLED, TRANSFERRED_TO). The website draws them with **Cytoscape.js**. Results are capped (for example 200 nodes) and the hop limit is 4, so the picture stays readable.
Why this choice	In SQL a 3-hop question needs a chain of table joins; in Neo4j it is one short query. **Alternatives:** Memgraph (similar, in-memory), PostgreSQL recursive queries (harder to read), NetworkX in memory (nothing is saved when it stops). For drawing: D3.js needs much more code; vis-network is simpler but gives less control.

```
MATCH path = (p:Person {name: 'Rahul Sharma'})-[*1..3]-(x)
RETURN path LIMIT 200
```

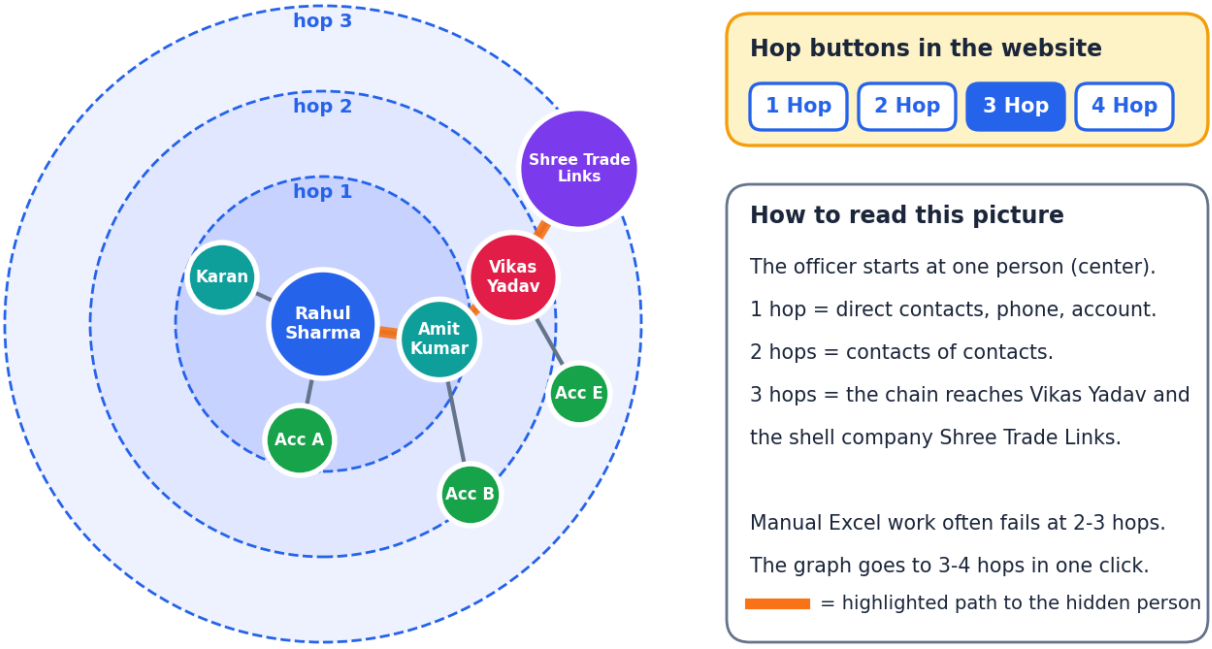


Figure 9: Hop buttons in the website. Three hops from Rahul reach the hidden person and the shell company.

HONEST WARNING

Showing too many nodes creates a 'hairball' that nobody can read. Always limit results, filter by type and amount, and let the officer expand step by step.

F8 Money Trail Explorer

HERO FEATURE

What it does	Pick an account, set an **amount filter** (for example ₹5 lakh and above) and a **hop limit**, and choose a direction: money in, money out or both. See the paths, totals, dates and transaction IDs.
Why it exists	Freezing money quickly is the most valuable action in a fraud case. Officers need to *see* the flow, not scroll through rows.
How it works	A Neo4j path query follows TRANSFERRED_TO links and checks the amount on every step. Pandas calculates totals and the time gap between money arriving and leaving. Patterns flagged for review: **many accounts into one** (fan-in), **one into many** (fan-out), **money leaving soon after arriving** (rapid pass-through), and **many amounts just below a limit** (splitting).
Why this choice	This was a teammate's idea and it makes the trail interactive. A static Sankey chart looks pretty but cannot be filtered. **Crypto wallets** are stored as nodes only; we do not claim crypto tracing in the first version.

```
MATCH path = (a:Account {number: '1234567890'})-[:TRANSFERRED_TO*1..3]->(b:Account)
WHERE ALL(r IN relationships(path) WHERE r.amount >= 300000)
RETURN path
```

TECHNICAL NOTE

In Neo4j a variable-length relationship returns a **list** of relationships, so a condition on every step must use **ALL(...)**. Writing *r.amount* directly on such a path is a common mistake and will not work.

Money Trail Explorer (example with filter = Rs 3 Lakh and above, 3 hops from Account A)

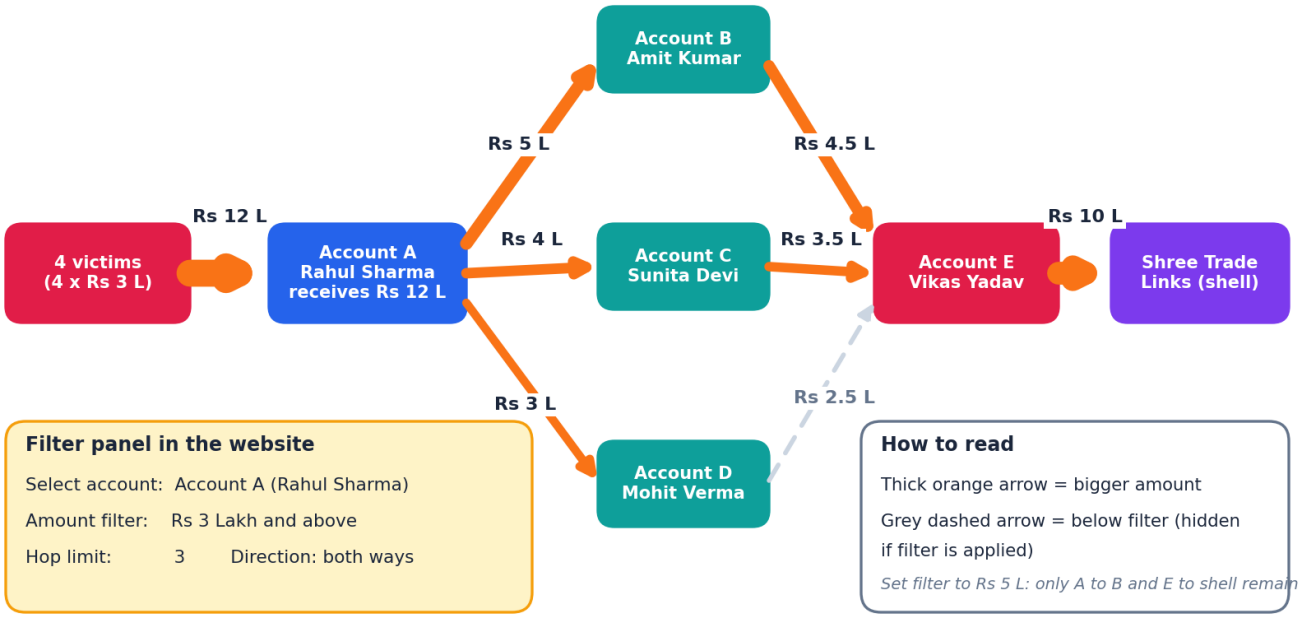


Figure 10: Money Trail Explorer for Case #101 with a filter of ₹3 lakh and above. Setting the filter to ₹5 lakh leaves only A to B and E to the shell company.

F9 Communication pattern detection

SUPPORT

What it does	Looks beyond *who called whom* to *how* they called: sudden spikes, very frequent contacts, bridge persons and suddenly new links.
Why it exists	Patterns give an early warning. In Case #101 the first alert appears on 03 Aug, six days before the first FIR.
How it works	Pandas counts calls per number per day. NetworkX measures bridges. Thresholds can be adjusted by the officer.
Why this choice	Simple rules can be explained in one sentence. Machine-learning anomaly detection (for example Isolation Forest) is a roadmap item.

Pattern	Plain meaning	How it is found
Sudden spike	A number's calls jump far above its own normal.	Compare each day with the number's normal (average plus 3 times the spread).
Very frequent contact	Two entities talk far more than typical pairs.	Rank pairs by calls per day and flag the outliers.
Bridge person	One person connects two otherwise separate groups.	Betweenness centrality plus group detection.
Sudden new connection	A new link appears and quickly becomes busy.	First-contact date plus call volume in the first days.

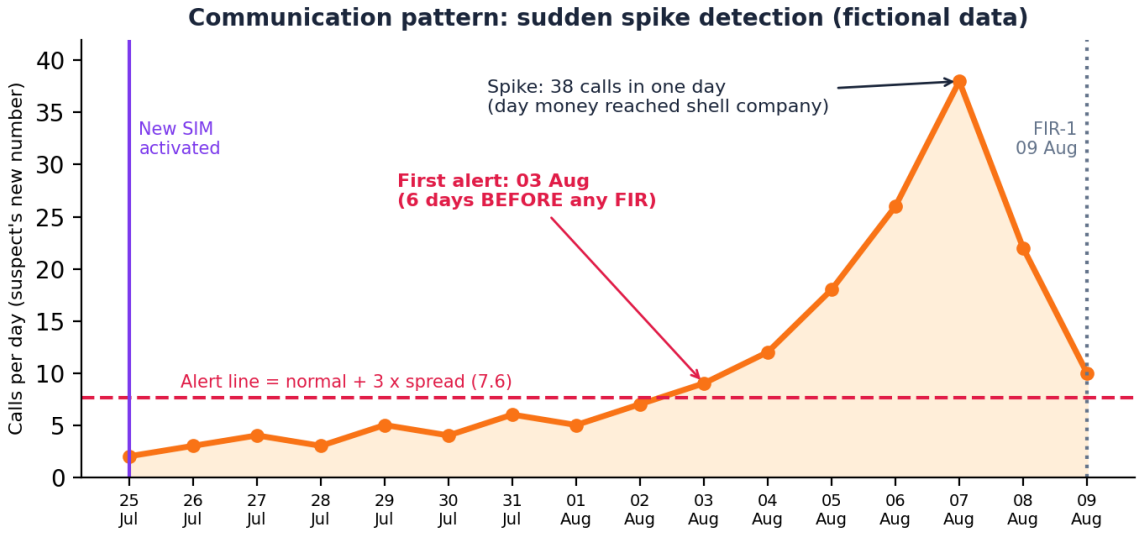


Figure 11: A new number's calls per day. The red line is the alert threshold.

HONEST TRUTH

A call centre or a shopkeeper also makes many calls. The system labels results **'Potential anomaly'**, shows the context and never says 'criminal'. Thresholds should be tuned on realistic data.

F10 Lead Advisor: 'Why is this person important?'

HERO FEATURE

What it does	A button on every person plus a list called **Suggested leads**. For each lead: priority score, reasons, evidence, matched patterns, **missing data**, suggested next steps and a confidence level. The officer chooses **Accept**, **Reject** or **Need more data**.
Why it exists	A picture is not enough. With thousands of nodes, officers need to know *where to look first* and *why*.
How it works	1. **Collect signals** from the graph: money, calls, network position, timing, links to several FIRs. 2. **Score each signal** with simple, visible maths. Weights are stored in a settings file. 3. **Compare with known crime patterns** (a small rule library, for example mule-account network). 4. **List missing data** that each pattern normally needs. 5. **Officer decides**; the decision is saved in the audit log.
Why this choice	A graph neural network or an ML classifier needs lots of labelled criminal data (we have none) and is a black box. Centrality alone does not explain anything. **Rules plus scores** are explainable and adjustable. An LLM may only rewrite computed facts into plain language; it must never add new facts.

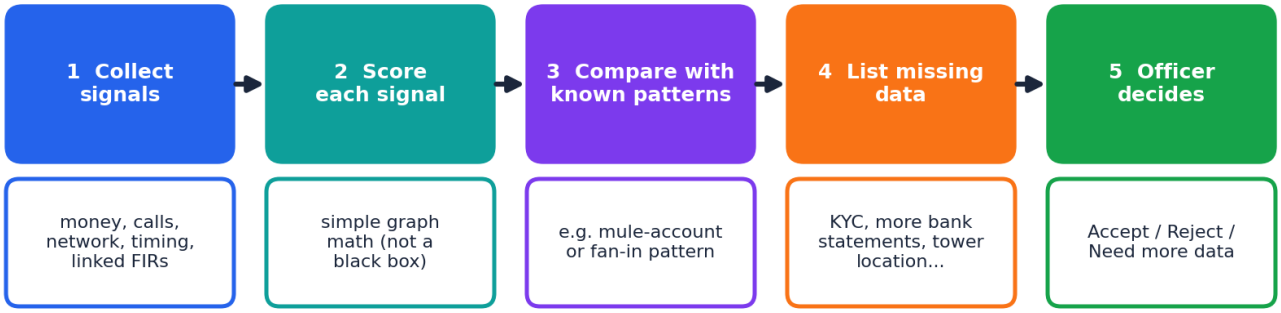
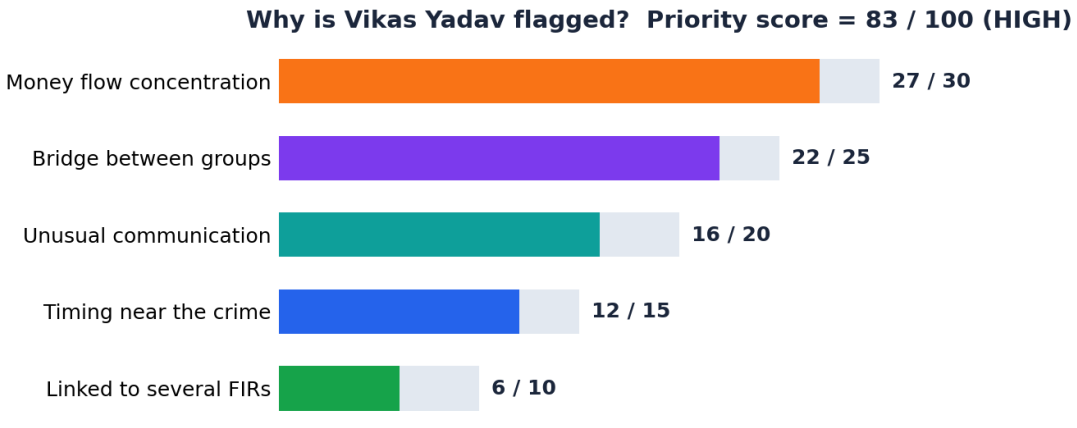


Figure 12: How a lead is produced and who has the last word.



Illustrative weights. Score = priority for human review, NOT proof of guilt.

Figure 13: The score is a sum of visible parts. Weights are illustrative and can be tuned.

SAMPLE LEAD CARD: Case #101

Lead 1: Vikas Yadav | Priority: **HIGH (83/100)** | Confidence: **Medium** (two data sources are missing)

Why flagged: only person who links both groups; ₹10 lakh moved from his account to a shell company; a new SIM made 38 calls on the day the money moved; the money from all 4 victims reaches his account (indirectly linked to all 3 FIRs).

Evidence: call-record rows (new SIM), bank transactions A to B to E to shell company, Master Profile links.

Pattern match: mule-account network (money collected from several accounts, then quickly passed on).

Missing data: KYC record for Account E; 15 more days of Account E statements; tower-location data for the new SIM; SIM purchase and verification details.

Suggested next steps: request KYC and SIM details; request tower data; consider asking the bank to review the account, following legal procedure. *Final decision: investigating officer.*

BIAS WARNING

Past criminal records: in the demo they are synthetic. In real life, access needs official authorisation. A past record must only be a small, clearly labelled signal, and it must never raise a score on its own, because that would be unfair bias against people who were already in the system.

F11 Timeline — SUPPORT

What it does	Shows events from all files in time order: FIR dates, calls, transfers and system alerts. Filter by person, type or date. Click an event to open its evidence.
Why it exists	Investigators and courts think in sequences. Order often shows cause and effect, for example a call spike right before a transfer.
How it works	Every source row becomes an event with a timestamp, type, linked entities and source. A timeline component draws them.
Why this choice	A calendar view is poor for dense events; a vertical or horizontal timeline reads naturally.

F12 'Why this link?' evidence button — SUPPORT

What it does	Click any line or circle in the graph and a small card shows what it connects, the **source file and page or row**, how it was found (rule, AI or officer), the confidence and who approved it.
Why it exists	Trust. In court and in front of judges the first question is always: *how do you know?*
How it works	Every node and relationship stores source_file, location, method, confidence and approved_by.
Why this choice	Cheap to build and very strong in a demo. It turns the graph from 'a nice picture' into 'evidence you can check'.

F13 Audit log with tamper check — SUPPORT

What it does	Records who did what, when and on which case or object: login, upload, approve, export and more.
Why it exists	Accountability. Seniors and courts can see exactly how the analysis was done.
How it works	Each entry stores user, time, action, object and a hash that includes the previous entry's hash. A 'Verify log' button re-checks the whole chain.
Why this choice	Our theme includes blockchain. A hash chain gives blockchain's key benefit (tamper evidence) without the complexity. A full Hyperledger or Ethereum network is heavy; a roadmap option is to anchor a daily summary hash on a permissioned blockchain.

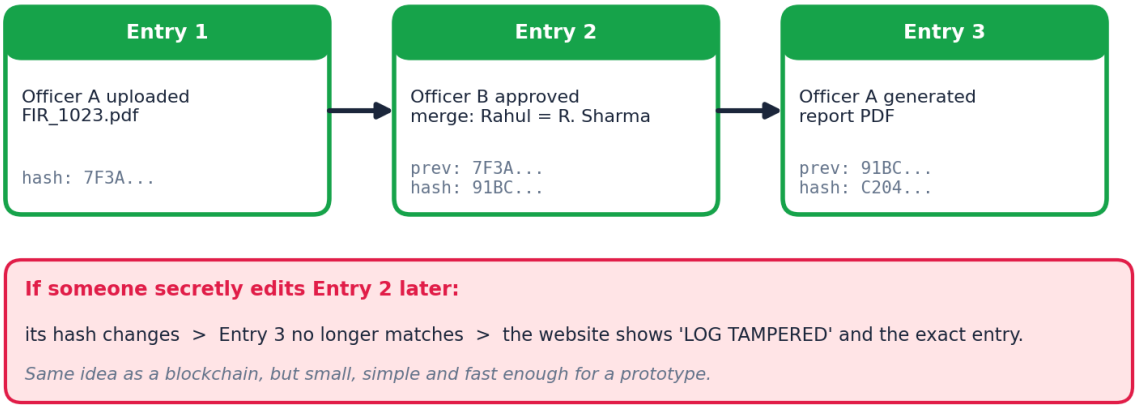


Figure 14: Each log entry carries the previous entry's hash. Editing one entry breaks every later one.

F14 Report generator — SUPPORT

What it does	One click creates a PDF: case details, entity summary, network summary with a graph picture, money trails, timeline, evidence references with file fingerprints, notes and lead decisions.
Why it exists	Officers must show their work to seniors and courts. This saves hours of typing and screenshots.
How it works	Python (ReportLab or WeasyPrint) fills a template with data from Neo4j and PostgreSQL. Every AI-generated item is labelled 'Lead for review'.
Why this choice	The alternative is pasting screenshots into Word by hand.

F15 Five small extras we suggest adding — EXTRA

Extra	What it does	Why it helps	Effort
Cross-case alert	When a new file arrives and a phone or account already exists in other cases, show an alert.	Catches repeat offenders instantly and solves the 'many FIRs' problem.	Low
Data quality check	After upload, list broken rows, invalid phone numbers and duplicates.	Real data is messy; this shows robustness.	Low
Lead decision log	Accept or Reject notes are saved and shown in reports.	Shows human-in-the-loop in action.	Low
Saved views	Save a graph state (for example Account A, ₹3 lakh and above, 3 hops) and reopen or share it.	Saves repeated work; great in a demo.	Medium

Extra	What it does	Why it helps	Effort
Plain-English summary	A few sentences at the top of each profile, written from computed facts using templates.	Beginners understand a profile in 10 seconds.	Low

5 How It Is Built

Architecture, data model, tools and why each tool was chosen

The system has **five layers**. Each layer has one job, so team members can work in parallel. The website talks to a small API layer; the API layer talks to a Python AI service and to two databases.

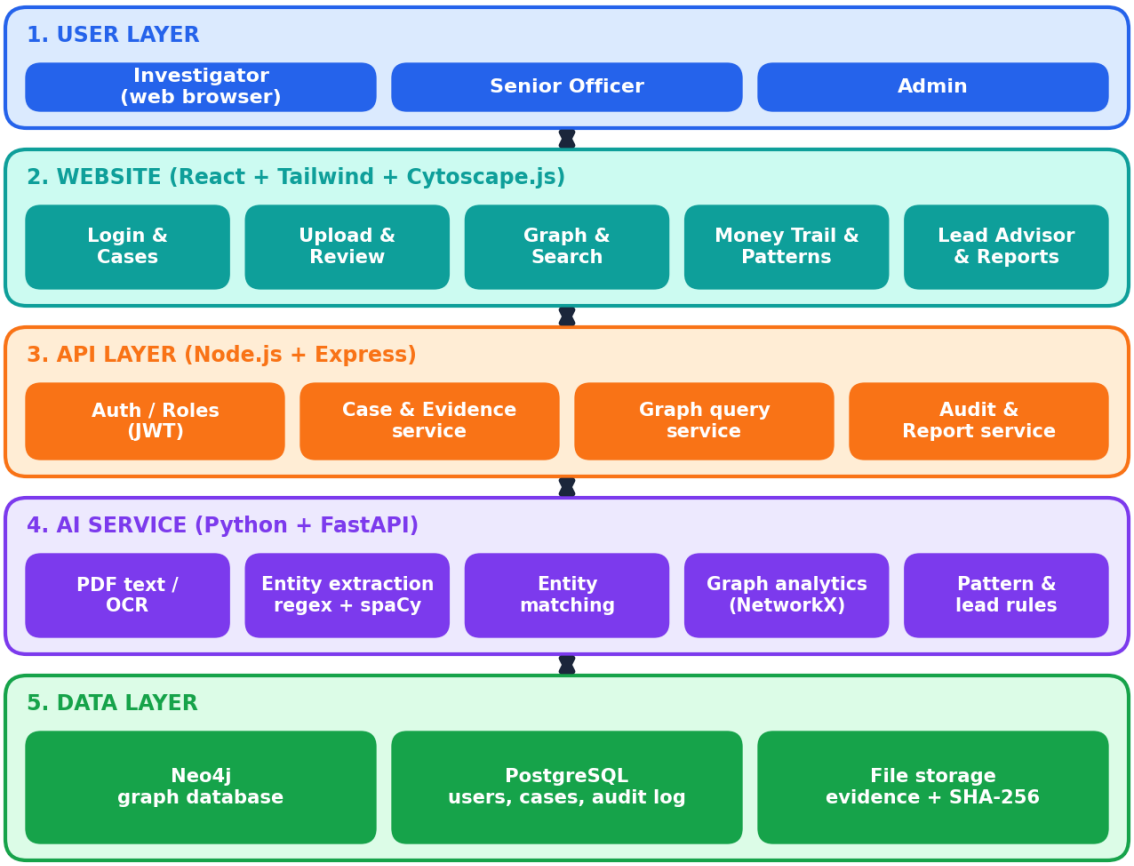
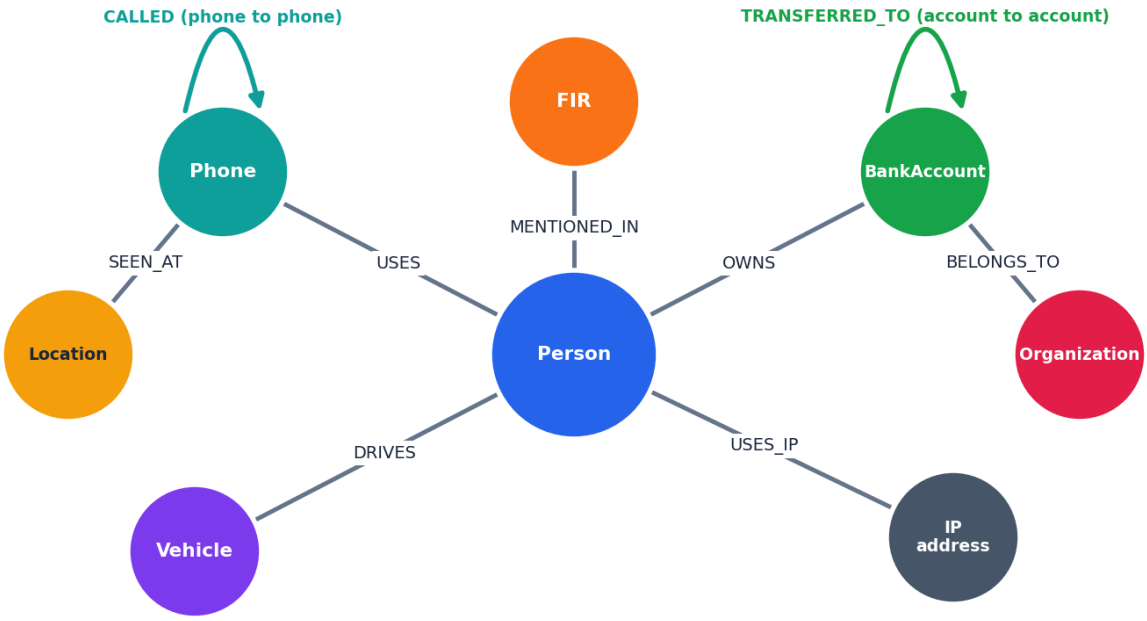


Figure 15: The five layers of Case Connect AI.

The graph data model

A graph has **nodes** (things: a person, a phone, an account) and **relationships** (links: uses, owns, called, transferred money). Every node and relationship also remembers **where it came from**.



Every node and every line also stores its SOURCE (file name + page/row) for the 'Why this link?' button.

Figure 16: The graph model. Phones link to phones by calls; accounts link to accounts by transfers.

Tools we will use, and why

Layer	Tool	What it does here	Why chosen	Alternatives
Website	React + Tailwind CSS	Pages, forms, tables	Huge community; AI coding tools write it well	Vue, Angular
Graph drawing	Cytoscape.js	Interactive network, layouts, click events	Built for network graphs	D3.js (more code), vis-network (less control)
API layer	Node.js + Express	Login, cases, uploads; talks to AI service	JavaScript, so the web team can build it	FastAPI only (fewest parts; best fallback)
AI service	Python + FastAPI	PDF reading, extraction, matching, analytics	Python has the best data and NLP libraries	Flask
Graph database	Neo4j (Cypher)	Stores entities and links; multi-hop queries	Most popular graph DB; free options; lots of learning material	Memgraph, PostgreSQL recursive queries
Normal database	PostgreSQL	Users, cases, permissions, audit log	Reliable, free, standard	MongoDB or SQLite for a quick prototype
PDF reading	PyMuPDF / pdfplumber	Text from PDFs	Fast and free	Tesseract OCR for scans (phase 2)
Extraction	spaCy + regex	Names, places; fixed patterns	Free, offline, explainable	LLM (flexible but may invent facts)
Matching	rapidfuzz	Name similarity (Jaro-Winkler)	Fast and simple	Machine-learning matchers
Data work	pandas	Read CSV or Excel; group by day; time gaps	The standard tool	-
Graph maths	NetworkX	Centrality, bridges, groups	Easy Python; no plugin setup	Neo4j Graph Data Science
Security	bcrypt, JWT, SHA-256, HTTPS	Passwords, tokens, fingerprints	Standard and proven	Never write your own
Reports	ReportLab / WeasyPrint	Creates the PDF report	Python-based	Word export
Test data	Python Faker (en_IN)	Fake Indian names, phones, addresses	Realistic, safe, free	Writing by hand
Team tools	Git + GitHub, VS Code, .env files	Code history, editing, secrets	Standard; commits prove your work	-

Key decisions: why this and not that

Decision	We chose	We rejected	Reason
Merging identities	Officer approval	Automatic merge above a score	A wrong merge misleads the whole investigation.
Explaining flags	Rules and visible scores	Graph neural network or black box	No labelled criminal data; every flag must be explainable to a court.
Evidence integrity	SHA-256 + hash-chained audit log	Full blockchain network	Same tamper-evidence, far less complexity and delay.
Reading documents	Regex + spaCy	LLM only	LLMs can invent facts, cost money and may send data outside.

Decision	We chose	We rejected	Reason
Graph scores	NetworkX in Python	Neo4j Graph Data Science plugin	Much easier to set up; enough for a prototype.
Data for demo	Synthetic data	Real or scraped data	Real investigation data is restricted; privacy and legal risk.
Hop limit	Maximum 4, results capped	Unlimited	Keeps the picture readable and the queries fast.
Wording	'Lead for review', 'Potential anomaly'	'Suspect', 'Criminal'	Fairness and legal safety; the system is not a judge.

IF TIME IS SHORT

The team presentation lists Node.js, FastAPI, Neo4j and PostgreSQL. If time is short, the safest simplification is to let **FastAPI** do both jobs (API and AI) and skip Node.js. Fewer moving parts means fewer bugs. If you do change the stack, update the slides so the demo matches what you present.

6 What Makes It Unique

How we compare with what already exists in the world

Several strong products already exist. The honest question is: **what is our place next to them?** We are not trying to beat them in every way. We are focused on one gap: **Indian multi-station evidence (FIR + call records + bank data), explained clearly, with a human in control, at low cost.**

Existing option	What it is	Where it is strong	The gap we target
IBM i2 Analyst's Notebook	Long-established link-analysis software used in investigations	Powerful charting for trained analysts	Commercial enterprise licence; reading Indian FIR text and explaining leads is not its focus (from public descriptions).
Palantir Gotham	Large intelligence and defence data platform	Integrates huge data sources at scale	Enterprise scale and cost; closed system; out of reach for a small unit or a student prototype.
Chainalysis / Elliptic	Cryptocurrency tracing tools	World-class crypto flow tracing	Crypto only; do not read FIRs or Indian call and bank records.
Maltego	Link-analysis tool, often used for online information	Gathering and charting online data	Focus is online information, not case files like FIRs and CDRs.
CCTNS	National network of police stations for FIR and crime records	Connects every police station; record keeping and sharing	Mainly record keeping and sharing; not an analysis tool for call and bank evidence (from public descriptions).
Excel and manual work	What many units use today	Free and familiar	Slow; breaks down after 2 to 3 hops; no proof of where facts came from.

Based on public descriptions of these products. They are mature and have many features not listed here. Our claim is **focus**, not superiority.

Our 8 differentiators

#	Differentiator	Why it matters
1	**Cross-FIR Master Profile**	Built for the real Indian situation: the same person appears in FIRs at many police stations.
2	**FIR + call records + bank data in one graph**	No hopping between tools; connections appear automatically.

#	Differentiator	Why it matters
3	**Explainable 'Why is this person important?'**	Gives leads with reasons, evidence, missing data and next steps, not verdicts.
4	**Human-in-the-loop by design**	Every merge and lead is approved by an officer, which lowers the risk of AI mistakes.
5	**Interactive money trail**	Amount and hop filters turn a static chart into an investigation tool.
6	**Provenance on every link**	Source file, page and row for each fact; trust and court-readiness.
7	**Tamper-evident audit log**	Blockchain-style protection without blockchain-style cost.
8	**Open-source, low-cost stack**	Runs on a normal laptop or an office server; suitable for state units.

HONEST TRUTH

Do not claim 'better than IBM i2 or Palantir'. Judges will not believe it. Say: '*Those tools are excellent but expensive and general. We focus on Indian FIR, call and bank evidence, explainable leads and human control, and we can run on open-source tools.*' Also do not claim regional-language support until Hindi OCR is really built.

7 What to Build First

Priorities, so the team does not get lost

A smooth demo of a smaller product beats a half-working demo of a huge one. The chart below ranks every feature by **impact** (how much it helps investigators and judges) and **effort** (how hard it is to build).

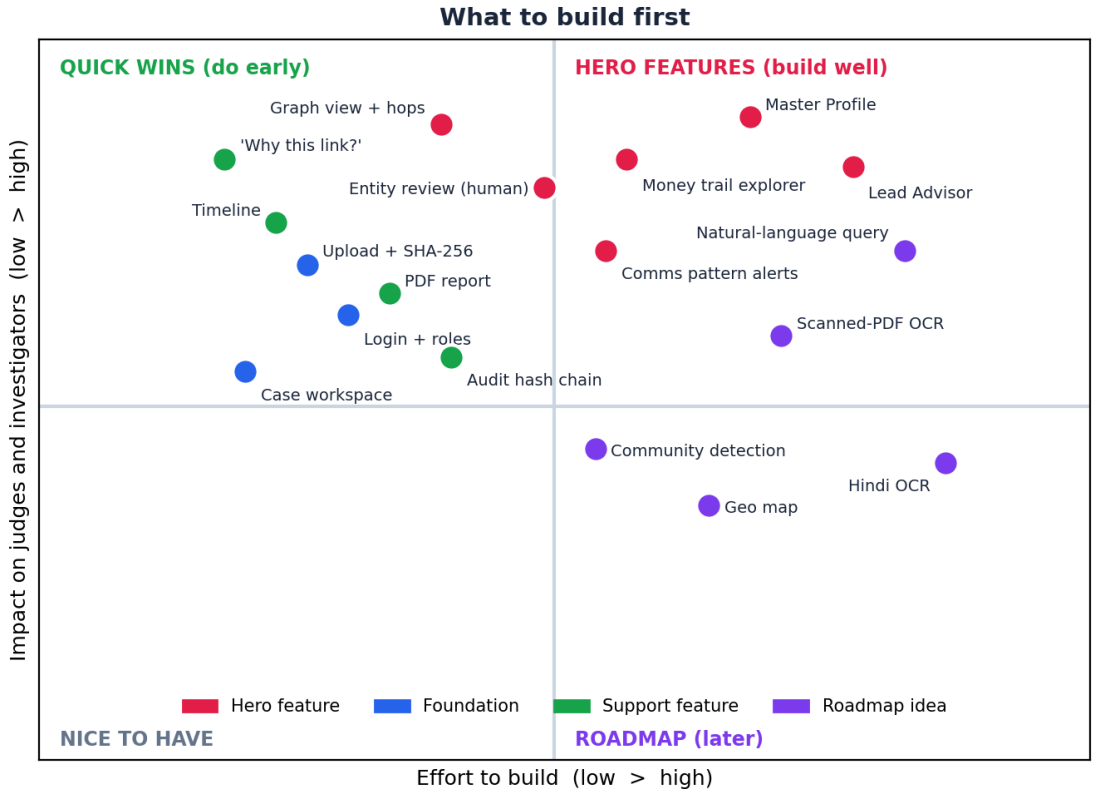


Figure 17: Build the red and blue items first, add the green ones next, and keep the purple ones for the roadmap.

Stage	What to build	Done when...
Stage 1: Foundation	Login and roles; cases; upload with SHA-256; graph in Neo4j; search with 1 to 4 hops	You can load Case #101 data and see the graph live.

Stage	What to build	Done when...
Stage 2: Intelligence	Extraction (regex + spaCy); entity review; Master Profile	A new FIR upload links to an existing person after approval.
Stage 3: Hero features	Money Trail Explorer; communication patterns; Lead Advisor	You can answer 'who is important and why' with evidence.
Stage 4: Trust and polish	'Why this link?'; timeline; audit log; PDF report; extras	The full demo runs without a single manual workaround.
Stage 5: Roadmap	OCR, geo map, community detection, natural-language questions	Only after everything above is stable.

DO NOT BUILD (YET)

- **Crypto tracing:** big, specialised and easy to overclaim.
- **Kafka, Hyperledger, distributed clusters:** a judge can ask hard questions about them, and they add nothing to a live demo.
- **Graph neural networks:** no labelled data, hard to explain.
- **Real criminal data:** not available and not legal to use without authorisation.
- **Perfect Hindi OCR:** keep for later; start with text PDFs.

8 Build Plan

What to learn, how to vibecode safely, timeline, team and test data

8.1 What you need to learn

AI can write most of the code, but **you must understand the key parts**. Otherwise you cannot fix bugs or answer judges. Learn these in this order:

Order	Skill	How deep	Free way to learn
1	Neo4j + Cypher	Basic to intermediate: nodes, relationships, MATCH, WHERE, paths	Neo4j GraphAcademy courses; Neo4j documentation
2	Graph ideas	Hops, centrality, bridges (the idea, not the maths)	NetworkX documentation; short videos
3	Cytoscape.js	Basic: add nodes and edges, style, click events	Cytoscape.js documentation demos
4	API basics (Node or FastAPI)	What a request and response are; routes	Official tutorials of Express or FastAPI
5	Python + pandas	Read CSV, filter, group by	pandas documentation, free tutorials
6	NLP basics	Use ready-made spaCy models and regex; no training	spaCy free online course
7	Matching and transactions	Name similarity, simple rules, money flow	Learn while building F5 and F8
8	Login and audit	Hashing passwords, tokens, roles	Library docs (bcrypt, JWT)
9	Git and GitHub	Commit, branch, pull request	GitHub documentation
10	Debugging	Read the error, test small parts	Practice; ask AI to explain errors

You do **not** need to learn deep learning, neural-network maths, training large models, computer vision or advanced blockchain development.

8.2 Ten rules for safe vibecoding

1. **One small task at a time.** Not 'build the whole app'. Start with 'upload a CSV and create nodes in Neo4j'.
2. **Write the data format first** (the JSON each part sends and receives), then ask AI for code.
3. **Run and test every step** before asking for the next one.
4. **Commit to Git after every working step.** The commit history is proof that you built it.
5. **Ask AI to explain** every important piece in plain words. If you cannot explain it, you cannot defend it.
6. **Own the key logic:** matching rules, Cypher queries, score calculation. Let AI do the boring parts (layouts, boilerplate).
7. **Keep one source of truth** for settings (weights, thresholds) in a config file.
8. **Never put passwords or keys on GitHub.** Use a .env file and a .gitignore.
9. **Test with messy data** (typos, missing values), not only perfect data.
10. **Freeze features one week before the event** and only fix bugs after that.

8.3 Example 8-week timeline

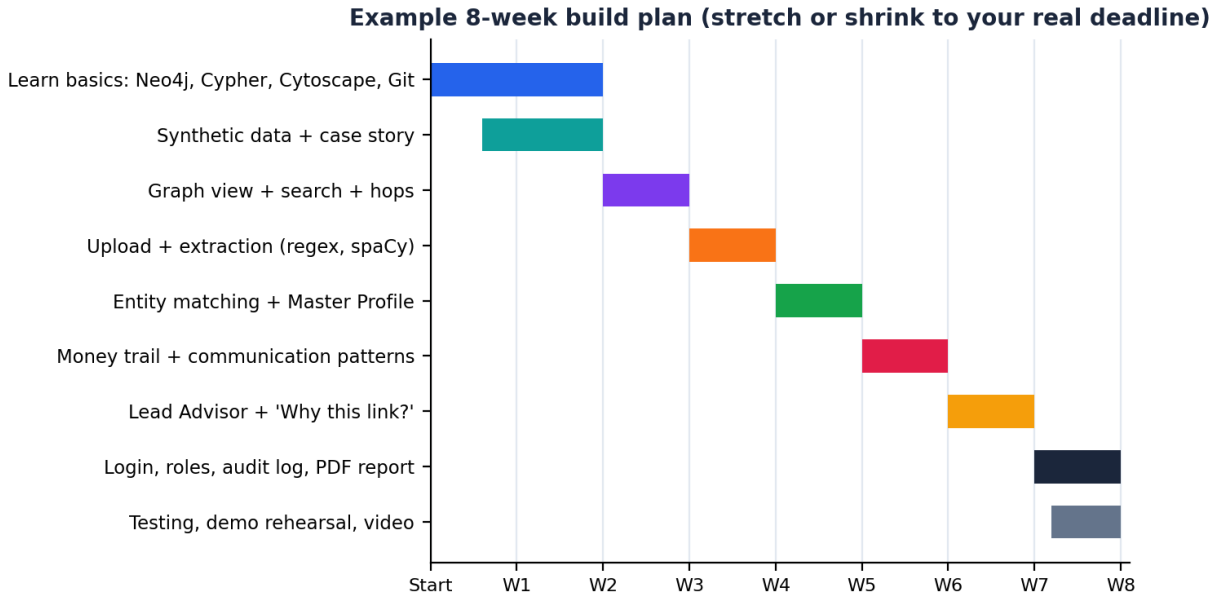


Figure 18: Adjust the weeks to your real deadline. Learning and data creation start together.

8.4 Team roles (for a team of six)

Role	Main job	Skills
Frontend	Pages, graph view, filters, Lead Advisor panel	React, Tailwind, Cytoscape.js
Backend	Login, roles, cases, uploads, audit log, API	Node.js + Express (or FastAPI), PostgreSQL
AI / NLP	PDF to text, extraction, entity matching	Python, regex, spaCy, rapidfuzz
Graph	Graph schema, Cypher queries, hop search, money trail	Neo4j, Cypher
Data / Analytics	Synthetic data, communication patterns, scores	Python, pandas, NetworkX, Faker
Integration and pitch	Connect everything, testing, slides, demo script, README	Git, Figma or slides, storytelling

8.5 The synthetic data plan (the most important early task)

Judges cannot see real police data, so a **good story in the data** is what makes the demo convincing. Build Case #101 first, then add noise.

Item	Starting size	Notes
People	About 15	6 in the core story, 9 as noise
Phone numbers	About 20	Includes Vikas's new SIM (activated 25 Jul)
Bank accounts	About 25	Accounts A to F as in Figure 10, plus noise
Call records	About 800 rows	Includes the call counts and the 38-call spike on 07 Aug
Bank rows	About 300	Includes the ₹12 lakh in, ₹10 lakh out, and the 2.5 lakh transfer below the filter
FIR text files	6	3 in the story (Rohini, Dwarka, Saket) plus 3 noise FIRs
Performance test later	About 10,000 nodes and 50,000 links	Use a generator to show the system stays fast

Hidden 'needles' to plant: Vikas Yadav as the bridge; a shell company; one person in 3 FIRs with name variants; two relatives sharing one phone (to test false merges); some typos, missing fields and duplicate rows (to test data checks).

8.6 Folder structure and main API endpoints

```
case-connect-ai/
  frontend/      React website
  backend/       Node.js API (login, cases, uploads, audit)
  ai-service/    Python FastAPI (extraction, matching, analytics)
  data/          synthetic data generator + sample files
  docs/          README, slides, demo script
  .env.example   settings template (never commit real .env)
```

Method	Endpoint	Purpose
POST	/auth/login	Sign in; returns a token
POST	/cases	Create a case
POST	/cases/:id/evidence	Upload a file; returns its SHA-256
GET	/cases/:id/entities	List extracted entities and match suggestions
POST	/matches/:id/approve	Approve (or reject, or undo) a merge
GET	/graph/search?q=&hops;=	Search and expand the graph
GET	/money-trail?account=&min;=&hops;=&dir;=	Money Trail Explorer
GET	/leads/:caseId	Suggested leads with scores
GET	/entities/:id/why	'Why is this person important?'
GET	/audit/verify	Check the audit hash chain
POST	/cases/:id/report	Generate the PDF report

8.7 Cypher cheat sheet (the 5 queries to know by heart)

```
// 1. Create two people and a call link
CREATE (a:Person {name:'Rahul Sharma'})-[:USES]->(p:Phone {number:'9876543210'})

// 2. Everything within 3 hops of Rahul
MATCH path = (:Person {name:'Rahul Sharma'})-[*1..3]-(x) RETURN path LIMIT 200

// 3. Money trail with an amount filter
MATCH path = (a:Account {number:'1234567890'})-[:TRANSFERRED_TO*1..3]->(b:Account)
WHERE ALL(r IN relationships(path) WHERE r.amount >= 500000)
RETURN path

// 4. Shortest connection between two people
MATCH p = shortestPath((a:Person {name:'Rahul Sharma'})-[*..6]-(b:Person {name:'Vikas Yadav'}))
RETURN p

// 5. Most connected people
MATCH (p:Person)-[r]-() RETURN p.name, count(r) AS links ORDER BY links DESC LIMIT 10
```

9 Demo and Judging

An 8-minute demo script, likely judge questions and live tests

9.1 The 8-minute demo script

Time	Screen	What to say
0:00 - 0:45	Problem slide	"Evidence sits in five rooms. Officers join it by hand. We built one connected picture."
0:45 - 1:30	Login, Case #101 pre-loaded	"This is a private case workspace. Roles control who can see it."
1:30 - 2:30	Upload a new FIR live	"AI reads the FIR. Every item has a confidence and a source page."
2:30 - 3:30	Entity review, then Master Profile	"The same person appears in three police stations. I approve, and one profile appears."
3:30 - 4:30	Graph search, 3 hops	"A name that is in no FIR appears: Vikas Yadav."
4:30 - 5:30	Money Trail, ₹3 lakh and above	"₹12 lakh comes in, splits, merges again, and ₹10 lakh leaves to a shell company."
5:30 - 6:30	Lead Advisor on Vikas	"Here is why. Click any link to see its source. Here is what data we still need."
6:30 - 7:15	Call-spike alert, timeline	"The alert fired six days before the first FIR."
7:15 - 7:45	PDF report, verify audit log	"One click for the report; the log proves nothing was edited."
7:45 - 8:00	Closing slide	"The system suggests. The officer decides."

Tip: pre-load Case #101 and cache any outside data. Never depend on live internet during the demo.

9.2 Questions judges may ask, with answers

Question	Short, honest answer
How do you handle aliases and many phone numbers?	Strong IDs (phone, account, vehicle) plus name similarity give a confidence level. The officer approves every merge, and merges can be undone.
What if relatives share a phone or address?	Shared phone or address alone is flagged as 'shared device or address', not merged. A similar name is also required, and the officer decides.
Does the AI decide who is guilty?	No. It produces *leads for review* with visible reasons, evidence and missing data. The officer decides.
Will this evidence stand in court?	We help with integrity: SHA-256 fingerprints, an audit log and source tracking for the certificate under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023. Admissibility is for the court.
Why Neo4j and not a normal database?	Multi-hop questions are short in Cypher and long in SQL. Normal data (users, cases, audit) still lives in PostgreSQL.
How does it scale to millions of call records?	The prototype is tested on thousands to tens of thousands of nodes. For scale we would add batch ingestion, indexes and a bigger graph setup. We do not claim more than we tested.
Where does your data come from?	Synthetic data for the demo. Real use needs authorised access and legal review.
How do you avoid bias and false flags?	Scores are visible sums of simple parts, weights are adjustable, past records have low weight, and wording is 'lead' or 'potential anomaly'.

Question	Short, honest answer
How are you different from IBM i2 or Palantir?	They are excellent and general. We focus on Indian FIR, call and bank evidence, cross-FIR profiles, explainable leads and human control, using open-source tools.
Why not use an LLM for everything?	LLMs can invent facts. We use rules and NER for extraction and use an LLM, if at all, only to reword facts we already computed.
How is data protected?	Roles, hashed passwords, HTTPS, audit log and fingerprints. A real deployment would be on a secure office server or government cloud with a security audit.
What if OCR or NER misses something?	Everything has a confidence level and goes to a review queue. Officers can add or fix entities by hand.
How would this work with CCTNS?	As an analysis layer, through authorised import and export adapters. That needs official approval and is on the roadmap.
Where is the blockchain?	In a hash-chained audit log: editing any entry breaks the chain. A roadmap step is to anchor a daily hash on a permissioned blockchain.

9.3 Live tests judges may try

Judge says	What you do
"Show entities within 3 hops of Suspect X with transfers above ₹5 lakh."	Open Money Trail, pick the account, set ₹5 lakh and 3 hops. Show the Cypher with ALL(...) if asked.
"How exactly are these two people connected?"	Use the shortest-path search; the path highlights and each link opens 'Why this link?'.
"Why was this person flagged?"	Open the Lead Advisor: score parts, evidence, missing data.
"What if you merged the wrong people?"	Click 'Undo merge'; show the audit log entry.
"Prove the file was not changed."	Re-run the SHA-256 check and the 'Verify log' button.

RED FLAGS TO AVOID

- Hard-coded graph results or screenshots instead of a live graph.
- Not being able to explain your matching rules.
- A demo that needs live internet or a slow first load.
- Starting the demo with scanned-PDF OCR, which is the flakiest part.
- Calling anyone a 'criminal' on screen.

10 Risks, Ethics and Law

The honest part: what can go wrong and how we handle it

#	Risk	What could happen	Our answer
1	Wrong merge of two people	Investigation goes in the wrong direction	Officer approval, confidence levels, 'Undo merge', audit log
2	False flag or bias	Unfair suspicion on an innocent person	'Lead for review' wording, visible scores, low weight for past records, adjustable thresholds
3	Synthetic data looks too perfect	Judges doubt the demo	Add typos, missing fields, duplicates and decoy people
4	Too many features	Nothing is finished or smooth	3 hero features, a feature freeze one week before the event
5	OCR or NER mistakes	Entities missed or wrong	Confidence, review queue, manual edit; demo with text PDFs
6	Unreadable or slow graph	A messy, slow demo	Result caps, filters, indexes, pre-loaded data
7	Learning curve (Neo4j, APIs)	Delays	Learn Neo4j first; small steps; pair up
8	Hackathon rules about prior work	Disqualification	Read each rulebook; keep the Git history; declare reused parts honestly
9	Privacy and law	Loss of trust, legal problems	Roles, audit log, synthetic data; real use needs authorised access and legal review
10	AI-written code nobody understands	Cannot answer judges or fix bugs	Ask AI to explain; own the key logic; test every step

Ethical principles we follow

- **Human decides.** The system never takes legal or enforcement action.
- **Leads, not verdicts.** High importance in a graph does not prove wrongdoing.
- **Transparency.** Every flag shows its reasons and sources.
- **Fairness.** Past records are a small, labelled signal; scores are reviewed for bias.
- **Minimum data and strict access.** Only authorised officers see case data.

LEGAL NOTES

Legal notes (not legal advice). Electronic evidence in India is now handled under the **Bharatiya Sakshya Adhiniyam, 2023** (Section 63 covers electronic records and their certificate), which replaced the older Section 65B of the Indian Evidence Act. India's data-protection law (the Digital Personal Data Protection Act, 2023) and police data rules would also matter for a real deployment. Before any real use, get advice from a lawyer who knows these laws.

11 Future Roadmap

Where the project can grow after the hackathon

Stage	What we add	Notes
Now: prototype	The MVP in Part 7, on synthetic data	Smooth, honest demo
Next: pilot-ready	Hindi and regional OCR, scanned documents, per-case roles, saved views, community detection, geo map from tower locations	Needs real, authorised sample data
Then: scale	Batch ingestion queue, indexes, bigger graph setup, performance tests on millions of rows	Add Kafka or clusters only if tests show a need
Then: intelligence	Bigger pattern library, natural-language questions that show the generated Cypher, ML anomaly detection, learning from officer decisions (with care)	Keep explanations visible
Then: trust	Anchor the audit hash on a permissioned blockchain, single sign-on, on-premise or government cloud, a security audit	Required for real deployment
Then: integration	Authorised import and export with CCTNS-style systems	Needs formal approval

12 Glossary

Every technical word, in one plain line

Term	Meaning
FIR	First Information Report: the first written record of a crime made at a police station.
CDR	Call Detail Record: a telecom file listing who called whom, when and for how long.
KYC	Know Your Customer: identity records a bank keeps for an account holder.
Graph	A map of things (nodes) joined by links (relationships).
Node	One thing in the graph: a person, phone, account, vehicle.
Relationship (edge)	A link between two nodes, such as CALLED or TRANSFERRED_TO.
Hop	One step along a link. Two hops means a contact of a contact.
Entity	Any real-world thing the system tracks: person, phone, account, vehicle, place.
Entity resolution	Deciding whether two records describe the same real-world thing.
NER	Named Entity Recognition: AI that finds names, places and organisations in text.
Regex	A pattern rule for text, such as ten digits for a phone number.
OCR	Optical Character Recognition: turns a scanned picture of text into real text.
Centrality	A score for how connected or important a node is in the network.
Betweenness	How often a node sits on the shortest path between others: a 'bridge' score.
Community detection	Finding groups of nodes that are tightly connected.
Mule account	A bank account used to pass on money for someone else.
Shell company	A company that exists mainly on paper, often used to hide money.
Fan-in / fan-out	Many accounts sending to one account, or one account sending to many.
SHA-256	A fingerprint of a file; changes completely if the file changes.
Hash chain	Log entries that each include the previous entry's fingerprint.
RBAC	Role-Based Access Control: what you can do depends on your role.
JWT	A signed token the server gives after login to prove who you are.
Audit log	A record of who did what and when.
Provenance	The origin of a fact: which file, page and row it came from.
Cypher	The query language of Neo4j.
API	A set of web addresses that programs use to talk to each other.
Synthetic data	Fake but realistic data created for testing and demos.
Human-in-the-loop	A person must check or approve important AI results.
MVP	Minimum Viable Product: the smallest version that shows the idea working.
Vibecoding	Building software by describing what you want to an AI tool and guiding it step by step.

13 Final Checklist and Sources

Before you build, before you present

Before you start building

- Read the rulebook of SIH and of each other hackathon, especially the rule about prior or pre-built work.
- Write the Case #101 story and generate the synthetic data first.
- Install Neo4j, load Case #101 and run the 5 Cypher queries.
- Create a GitHub repository; commit after every working step.
- Agree on the JSON formats between modules before writing code.

Before you present

- The demo runs end to end on a freshly started laptop, without internet.
- You can explain the matching rules, the score parts and the Cypher query in your own words.
- Every screen uses 'lead' and 'potential anomaly', never 'criminal'.
- Slides match the real stack and the real features.
- The README, a 2-minute demo video and a backup of the data are ready.

The whole project in one line

Scattered data	Extraction	Matching (officer approves)	Knowledge graph	Analytics	Officer's decision

Sources and notes

- Team documents: SIH26189 skills guide, project architecture notes, the SIH problem analysis and the UNITx idea presentation.
- Ministry of Home Affairs, status of CCTNS implementation as on 01.02.2026 (Press Information Bureau document).
- Ministry of Home Affairs reply in the Rajya Sabha on 5 August 2026, reporting that all 17,762 police stations use CCTNS as of 1 June 2026 (as reported in the news).
- Bharatiya Sakshya Adhiniyam, 2023, Section 63 (electronic records).
- Product descriptions of IBM i2, Palantir Gotham, Chainalysis, Elliptic and Maltego are from public information and are summarised briefly. Check their current websites before quoting them.
- Tool documentation: Neo4j, Cytoscape.js, spaCy, FastAPI, pandas, NetworkX.

IMPORTANT NOTE

All scores, weights, confidence values and example numbers in this guide are **illustrative design choices** for a prototype on synthetic data. They are meant to be tuned and tested, not quoted as measured results.